



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Civil Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: B.E, I Civil Engineering

Course Code & Title: GE3151 & Problem Solving and Python Programming

Name of the Faculty member (s): Mrs.B.Yazhini AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit II /Operators
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen:** Virtual Lab
- **Justification:**

The Virtual Laboratory is an interactive environment for creating and conducting simulated experiments: a playground for experimentation. A virtual laboratory is an on screen simulator that learners use to test ideas and observe results. Since the principles of operators is mandatory for the programming language, hands on demonstration is given to the students through Virtual lab by using the following link
<https://python-iitk.vlabs.ac.in/exp/arithmetic-operations/simulation.html>

- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**

The course instructor explained the concept of the various operators. He opened the Virtual Lab using aforementioned link and demonstrates the various operators working to the students. At last 10 minutes, the course instructor chosen few students and instructed them to explain the various examples of operators by the students.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PSO 3
CO 1	2	2	2	1

(1 – Low

2 – Moderate

3 – High)

• PO / PSO mapped:

Innovative practice	PO 1	PO 2	PO 3	PSO 01
	3	2	1	2
Justification for correlation	Students could be able to apply basic theory and principles of operators	Students could be able to have the hands-on with operator on Virtual lab	Students could be able to recollect the basic principles of operators	Students could be able to knowledge about the operator to their peer group of people.

• Images / Screenshot of the practice:

Fig 1: Demonstration on operators

Fig:2 Demonstration on analysis of arithmetic operators

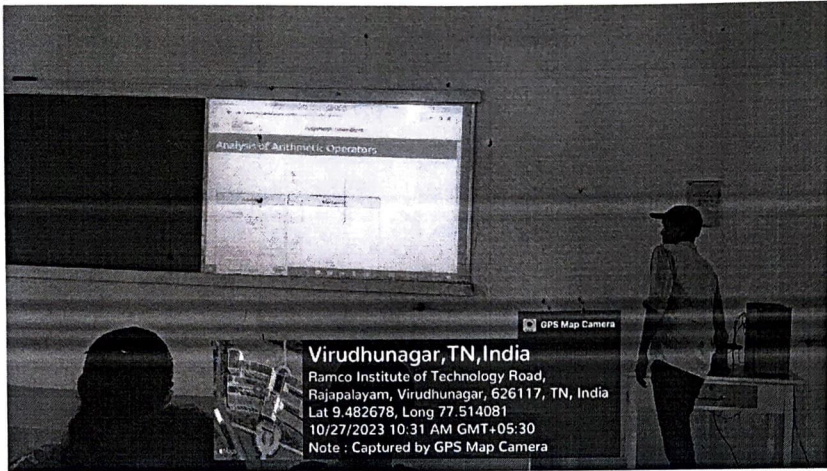


Fig 3. Students involved in demonstration activity

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students easily understand the working of operators and its corresponding code.
- Students said that the pretest and posttest is very useful to recollect the concepts.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students had understood the concepts of the various operators.
- Most of the students were actively participated and enjoyed once they answered correctly in pretest.

❖ **Challenges faced in implementation:**

- Some students only involved in the demonstration activity.
- The same activity will be planned in the laboratory session to make all the students familiar in the virtual environment.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/2021-2022/MGM_VII_Vlab_Demo.pdf
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/CS6801-MCAP-MGM.pdf>

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